

EmotionCam: A Camera-Based Interactive System for Real-Time Emotion Recognition and Sticker Generation

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Abstract—In our daily life, where texting involves a lot of humor finding the perfect stickers according to humor and expressions is very difficult. In this paper, we are introducing "EmotionCam", an interactive camera based system that detects our expressions and gives us a sticker based on it. We are using a CNN based model and have trained it on the FER2013 dataset, to be able to detect five different emotions: happy, sad, neutral, angry and surprised. The system first used webcam to detect our expressions and then system generates corresponding animal stickers. Those generated stickers are then stored as downloadable PNG image in a folder. Python libraries like TensorFlow for deep learning of model, and OpenCV, MediaPipe for expressions detections, were used, which was further tested on five different participants to check the accuracy of our system, to demonstrate how this system can bridge a gap, people face in daily life.

Index Terms—Facial Emotion Recognition, CNN, Human Computer Interaction, OpenCV, Deep Learning, Real-Time Detection

I. INTRODUCTION

A. Background of HCI and Multimodal Systems

Human-Computer Interaction (HCI) has progressed significantly from traditional text-based systems to more intelligent, natural, and user-centered environments. Human communication highly depends on non-verbal signals such as facial expressions, tone of voice, and body language, which help convey emotions and meaning beyond spoken words. To reduce the gap between human behavior and machine understanding, modern HCI systems increasingly use multimodal affective computing techniques. These systems combine facial expression recognition, speech analysis, and contextual awareness to enable machines to respond intelligently according to the emotional state of users.

B. Problem Motivation

Facial expressions play an important role in human non-verbal communication and strongly influence social interaction. Automated Facial Expression Recognition (FER) has therefore become an important research area in applications such as mental health monitoring, virtual learning environments, customer support systems, and driver safety systems.

However, accurately recognizing human emotions remains a difficult computer vision challenge due to intra-class variations, lighting changes, facial occlusions, pose variations, and differences among individuals.

C. Importance of the Proposed System

Facial Expression Recognition systems are essential for transforming traditional command-based systems into intelligent and context-aware interactive systems. Modern FER systems reduce dependence on manually engineered features and controlled environments, allowing them to operate more effectively in real-world conditions. Improving both recognition accuracy and computational efficiency is important for deploying emotion-aware systems on consumer devices and edge-computing platforms.

D. Research Gap

A detailed review of existing facial expression recognition research highlights several limitations in current systems.

1) Static Context and Spatial Alignment Gap:

Many traditional FER models mainly focus on localized facial regions while ignoring broader contextual information that may influence emotion understanding. In addition, conventional Convolutional Neural Networks (CNNs) may lose important spatial information during pooling and downsampling operations. This reduces the model's ability to detect subtle facial expressions and micro-expressions accurately.

2) Accuracy and Computational Trade-Off:

Modern deep learning models that use attention mechanisms, transformers, and depth-wise separable convolutions often achieve high accuracy but require large computational resources. These systems become difficult to deploy on low-resource devices. Lightweight FER models provide faster inference speed but often experience reduced accuracy under real-world conditions such as poor lighting, pose changes, and partial facial occlusion.

E. Proposed Contribution

The main contributions of this paper are:

- A real-time CNN-based facial emotion recognition system trained on the FER2013 dataset and integrated with a live webcam interface using OpenCV and TensorFlow.
- An interactive sticker generation system that automatically generates animal-based stickers according to the detected facial emotion and the user-selected animal theme.
- A usability evaluation demonstrating the system's ease of use, response speed, and user experience across multiple participants.

II. RELATED WORK

Talegaonkar et al. [1] implemented a CNN trained on FER-2013 for real-time emotion classification, achieving 79.89% training and 60.12% test accuracy, though performance degraded under poor lighting and real-world variations.

Cultrera et al. [2] explored spatio-temporal transformers for facial action unit classification using event cameras, introducing the FACEMORPHIC dataset, though the model is computationally complex and requires high-quality data.

Haq et al. [3] extended real-time recognition using MobileNet-V1 on FER-2013, reporting 97.9% training accuracy, but noted misclassification between visually similar emotions in practical scenarios.

Chindiyababy et al. [4] proposed a lightweight multi-modal CNN framework incorporating Grad-CAM explainability, trained across FER-2013, AffectNet, and CK+, outperforming traditional SVM and KNN classifiers.

Yang et al. [5] improved real-time responsiveness through edge computing, offloading AU-based emotion detection to local devices such as Raspberry Pi to reduce network latency, though accuracy declined in complex real-world conditions.

Gunes et al. [6] proposed a vision-based multimodal framework integrating facial and body gesture recognition across multiple pipeline stages, though no dedicated evaluation dataset was used.

Tu et al. [7] demonstrated face and hand gesture-based device control using basic image processing and a neural network, achieving 92% face detection and 93.6% gesture recognition accuracy, though performance remained sensitive to lighting and occlusion.

Iachello and Hong [8] synthesized three decades of HCI and CSCW privacy research into a unified framework, identifying five grand challenges for future privacy-aware system design, with focus on the end-user perspective.

Wei et al. [9] surveyed personalized image generation techniques using GANs and diffusion models over the past decade, highlighting high computational cost and inconsistency in personalized outputs as key open challenges.

Jana et al. [10] proposed DARKLY, a privacy protection framework for camera-based perceptual applications that selectively hides sensitive image data while preserving system functionality, evaluated across 20 applications.

Jacob and Stenger [11] applied transformer architecture with attention mechanisms and multi-task learning for facial action unit detection, achieving strong F1-scores on BP4D, DISFA,

and EmotioNet datasets, though the model demands significant computational resources.

Kolla [12] developed Emojify, a CNN-based real-time webcam system trained on FER-2013 to generate personalized emojis, evaluated through precision, recall, F1-score, and accuracy, though limited to a small set of emotion categories.

Xu et al. [13] combined CNN-based facial analysis with LSTM speech processing for multimodal real-time emotion recognition, demonstrating reduced task completion time and improved satisfaction across 30 participants, though multi-modal integration remains computationally complex.

Lemboye [14] examined user perceptions of facial recognition privacy through interviews with 15 participants, finding that most users had limited awareness of how their personal data was collected, though the study was constrained by its small sample size.

Alipour et al. [15] introduced Emoticontrol, a reinforcement learning-based system that dynamically adapts UI elements such as font size, color, and layout based on detected emotions, demonstrating improved user experience and faster decision-making in an emergency training mobile application, though effectiveness depends heavily on accurate emotion detection.

III. PROPOSED METHODOLOGY

In the emotioncam system, the webcam is used to take video input of the user. After that the frames that are captured are processed using HAAR Cascade Classifier to detect human faces. Once the face detection occurs, important facial features are extracted and passed to a CNN model. In the next step the emotion detection occurs which is performed by CNN. The five emotions emotioncam is dealing with are happy, sad, neutral, angry, and surprised. The third step is the selection of animals whose sticker user wants to generate. After animal selection, in the final step the system generates stickers based on the selected animal and the detected emotion which is then saved as a PNG image in a folder from where the user can share it in chats.

A. System Overview

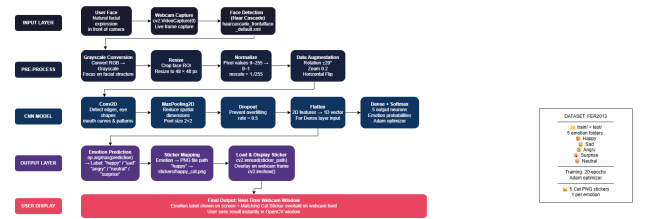


Fig. 1. System Architecture

B. Dataset Description

For training the emotioncam system FER-2013 dataset is used for facial emotion detection. The dataset consists of grayscale images which are 48x48 pixels in size. For this project we used 5 emotion classes

- Happy

- Sad
- Neutral
- Angry
- Surprised

The dataset consists of training and validation sets using approximately 80:20 ratio. For training the system 24,176 images are used and 6,043 images are used for validation and testing. Data augmentation techniques are to enhance the performance of the system such as image rotation, zooming, horizontal flipping using ImageDataGenerator function to make the system more generalized and reduce overfitting.

TABLE I
DATASET INFORMATION

Dataset	FER2013
Classes	5
Training Images	24176
Validation Images	6043
Image Size	48x48

C. Face Detection

The HAAR Cascade Classifier is used for real time face detection, it is a ML based object detection method provided by OpenCV. It detects human faces by extracting important face features like eyes, mouth and nose. To locate human faces we used a webcam which captures video frames and HAAR Cascade processes each frame. The image is converted into grayscale format after detecting faces. Before passing to the CNN model for emotion detection, the image is resized to 48x48 pixels.

D. CNN Model

The CNN model plays a role in face emotion detection as is used for image classification and feature extraction. The developed CNN model has multiple layers such as convolution layers, max-pooling layers, flatten layers, dense layers, and dropout layers. The first convolution layer uses 32 filters with ReLU activation function to extract basic features, max-pooling is used for reducing image dimensions and complexity. For extracting deep facial features second convolution layer has 65 filter is used followed by another max-pooling layer. The extracted features are flattened into 1D vectors and passed to a dense layer for classification. The dropout layer reduces the overfitting while training. The output layer used the Softmax activation to recognize facial expressions into five emotion categories (happy, sad, angry, neutral, and surprised).

IV. RESULTS AND DISCUSSION

The emotioncam system is successfully trained and tested for real-time facial emotion detection and sticker generation. The model showed gradual improvement in training and validation accuracy. The final model achieved 62.25% validation accuracy after 20 training epochs. The accuracy and loss graph showed that model learned facial features correctly during training. The system was also tested using webcam, HAAR

Cascade detected the faces and CNN model predicted emotions correctly. After that the system showed animal options to the user, after selecting animal, the system generated related sticker and save it as a PNG image in a folder. The system provides smooth and interactive user experience.

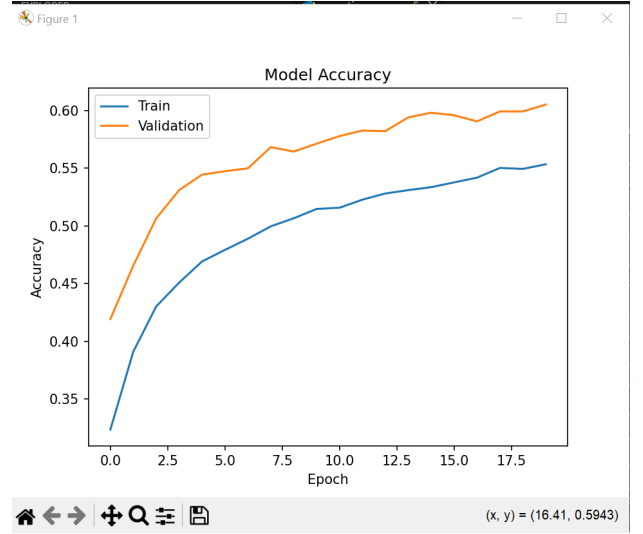


Fig. 2. Training and Validation Accuracy

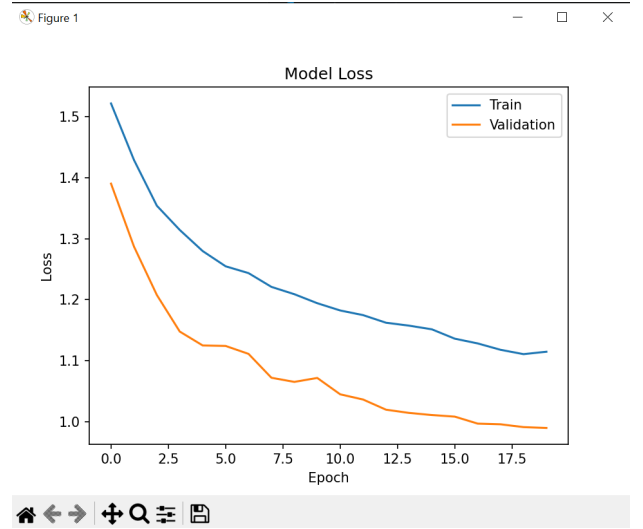


Fig. 3. Training and Validation Loss

TABLE II
MODEL PERFORMANCE

Metric	Value
Validation Accuracy	62.25%
Epochs	20
Optimizer	Adam
Loss Function	Categorical Crossentropy
Batch size	64

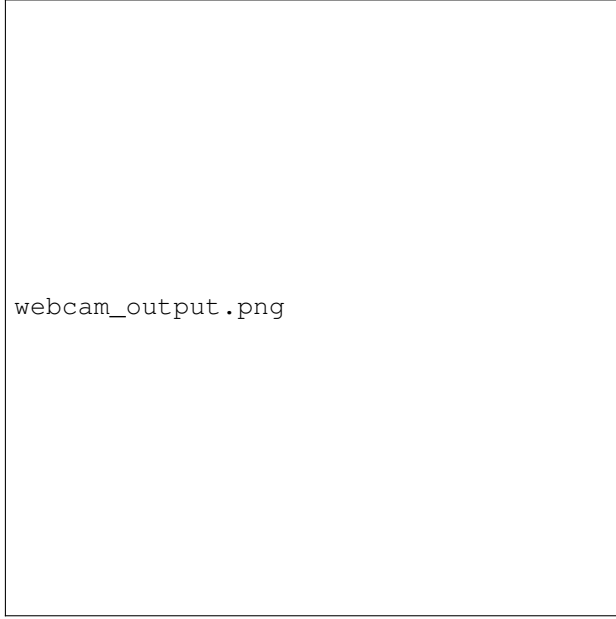


Fig. 4. Real-Time Emotion Detection

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=====
WEBCAM STARTED
Show Your Emotion Clearly
Hold Expression For Few Seconds
=====

=====
Emotion Detected: neutral
=====

=====
1 = Cat Sticker
2 = Monkey Sticker
3 = Panda Sticker
4 = Dog Sticker
5 = Detect New Emotion
6 = Exit
=====
Enter Choice: 1

CAT sticker generated!
Saved in folder: generated_stickers

```

Fig. 5. Console-Based User Interaction



Fig. 6. Generated Animal Sticker Output

V. USABILITY EVALUATION

The emotioncam system was tested by 5 different users to evaluate usability and user experience. The users were asked to perform different facial expressions and select animal stickers based on detected emotions. Most users found the system easy to use and interactive. The emotion detection worked smoothly and stickers were generated successfully. Users also liked personalized animal sticker generation feature because it made chatting and communication more fun.

TABLE III
USER FEEDBACK

User	Ease of Use	Response Speed	User Experience
User 1	4.5	4.2	4.8
User 2	4.3	4.0	4.6
User 3	3.8	3.9	4.1
User 4	4.7	4.4	4
User 5	4.5	4.3	3.7

VI. CONCLUSION

In this paper an iterative real-time emotion detection system called Emotioncam was proposed and implemented. The system uses a webcam to detect facial expressions and classify emotions using CNN model. After that the system allow users to select an animal theme and generated corresponding stickers. The dataset we used was FER-2013 and achieved 62.25% validation accuracy. The system is a combination of emotion recognition, real-time interaction, and sticker generation to improve human computer interaction. The usability evaluation also showed that users found the system simple and interactive.

VII. FUTURE WORK

In the future the system can be improved by increasing the number emotion classes and adding more animal stickers. Advanced learning models can be used to improve accuracy. Additional features include mobile application support.

VIII. COMPARATIVE ANALYSIS

Table IV summarizes and compares all fifteen reviewed papers across key dimensions.

TABLE IV
COMPARATIVE ANALYSIS OF REVIEWED LITERATURE

Paper	Method	Evaluation	Dataset	Limitation
[1]	CNN-based FER	Accuracy analysis	FER-2013	Lighting sensitivity
[2]	Transformer-based FER	Accuracy comparison	FACEMORPHIC	Complex model
[3]	MobileNet CNN	Accuracy testing	FER-2013	Emotion similarity issue
[4]	Explainable CNN	Precision and recall	AffectNet, CK+	Real-world variations
[5]	AU-based FER	Runtime analysis	RAF-DB	Reduced real-time accuracy
[6]	Face + gesture recognition	System analysis	Framework-based	Limited datasets
[7]	Gesture-based control	Accuracy testing	Real-time input	Lighting issues
[8]	Privacy framework	Qualitative analysis	Research literature	User-focused only
[9]	GAN image generation	Comparative study	Multiple studies	High complexity
[10]	Privacy protection system	Functionality testing	OpenCV apps	Complex scenarios
[11]	Transformer AU detection	F1-score analysis	BP4D, DISFA	High computation cost
[12]	Emoji generation	Accuracy testing	FER-2013	Limited emotion classes
[13]	CNN + LSTM	User comparison	FER-2013, RAVDESS	Integration complexity
[14]	Privacy perception study	Interviews	15 participants	Small sample size
[15]	Adaptive UI system	User feedback	Mobile app	Depends on emotion accuracy

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